

Zero-Shot vs Few-Shot Prompting Showdown

Week 1 · Generative AI & Prompt Engineering — NeuroFive Solutions Internship

1. Objective

This task explores how prompt design affects the accuracy and reliability of large language models on a real classification problem. The same set of 10 customer support messages was classified twice by each AI tool: first with a zero-shot prompt (instruction only, no examples), then with a few-shot prompt (instruction plus three labeled examples). Accuracy was measured against a manually prepared answer key. Tools tested: ChatGPT, Claude, and Gemini.

2. The Dataset: 10 Customer Support Messages

#	Message	True Label
1	My order arrived two weeks late and the box was crushed.	Complaint
2	Do you ship to Canada?	Question
3	Honestly the best customer service I've ever experienced, thank you!	Praise
4	I was charged twice for the same order and nobody is responding to my emails.	Complaint
5	How do I reset my account password?	Question
6	The new app update is so smooth, great job team!	Praise
7	What's your return policy for opened items?	Question
8	This is the third time my subscription renewed without warning. Unacceptable.	Complaint
9	Can I change my delivery address after placing the order?	Question
10	Shoutout to Sarah from support — she fixed my issue in minutes!	Praise

3. Zero-Shot Prompt

Classify each of the following customer support messages as "Complaint", "Question", or "Praise". Output only the message number and label.

[10 messages listed above]

4. Few-Shot Prompt

Classify each customer support message as "Complaint", "Question", or "Praise".

Examples:

Message: "My package never arrived and I want a refund now." → Complaint

Message: "Is there a student discount available?" → Question

Message: "You guys always go above and beyond, love this company!" → Praise

Now classify these. Output only the message number and label.

[10 messages listed above]

5. Results

Tool	Zero-Shot Accuracy	Few-Shot Accuracy
ChatGPT	10 / 10 (100%)	10 / 10 (100%)
Claude	10 / 10 (100%)	10 / 10 (100%)
Gemini	10 / 10 (100%)	10 / 10 (100%)

All three tools were tested with identical prompts in fresh chat sessions. Every tool returned the same labels in both settings: 1-Complaint, 2-Question, 3-Praise, 4-Complaint, 5-Question, 6-Praise, 7-Question, 8-Complaint, 9-Question, 10-Praise — all matching the answer key.

6. Analysis & Takeaway

All three models achieved perfect accuracy in both zero-shot and few-shot settings, so on this particular dataset the few-shot examples did not change the final scores. This tells us something important: modern frontier LLMs (ChatGPT, Claude, Gemini) have become strong enough that simple, clearly-worded three-way classification no longer needs examples to work. However, the few-shot prompt still provided real benefits — it guaranteed a consistent output format across all three tools and made the category boundaries explicit, which reduces the risk of format drift or hesitation on borderline messages. The key lesson is that few-shot prompting matters most when the task is ambiguous or the data is messy: real customer messages often mix a complaint with a question in one sentence, and that is exactly where labeled examples give the model the decision boundary it needs. For easy, well-defined tasks, zero-shot is sufficient; as task difficulty rises, examples become the cheapest and most effective way to boost reliability.

7. Conclusion

Prompt wording controls model behavior. Even when accuracy is identical, guided prompts produce more predictable, better-formatted, and more trustworthy outputs — the foundation of every prompt engineering technique covered in the coming weeks.

8. Submission Checklist

- ✓ Zero-shot prompt written and tested in ChatGPT, Claude, and Gemini — Done
- ✓ Few-shot prompt written and tested in all three tools — Done
- ✓ Accuracy comparison table completed — Done
- ✓ Written takeaway (Section 6) — Done
- 2–3 min walkthrough video recorded and uploaded to LinkedIn (tag NeuroFive Solutions)